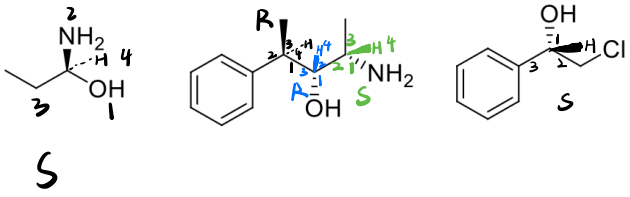
Question 1 (20 marks)

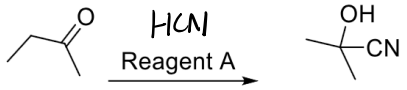
Assign the configuration (*R* or *S*) to each stereocentre in the molecules shown below. Clearly show how you have worked out your answer. (10 marks for the assignment, 10 marks for the justification).

Show the CIP priority rules for every single stereocenter

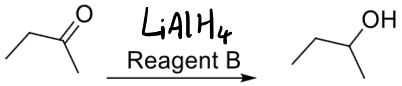


Question 2 (10 marks)

Propose suitable reagents and provide plausible mechanisms for the formation of compounds **2** and **4** (5 marks for each reaction, 2 for the reagents, 3 for the mechanism).



1 **2**

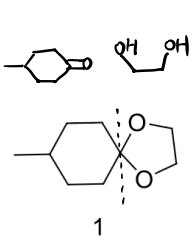


3 **4**

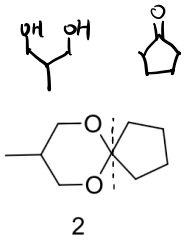
Answer:

Question 3 (10 marks)

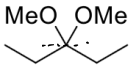
Suggest a combination of a carbonyl compound and an alcohol to produce each of acetals **1** to **4**.

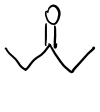


1

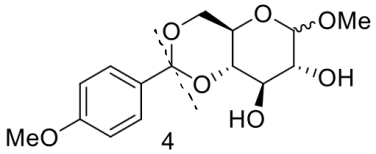


2



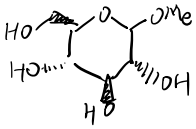


3



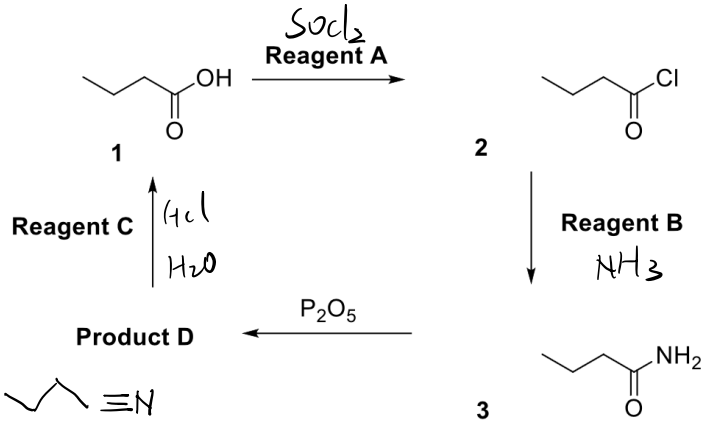
4





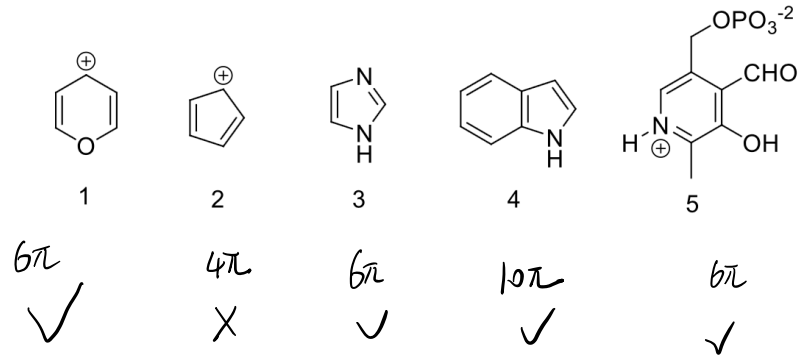
Question 4 (10 marks)

Identify the reagents **A, B, C** and the product **D**. No mechanisms required. (2.5 marks each)



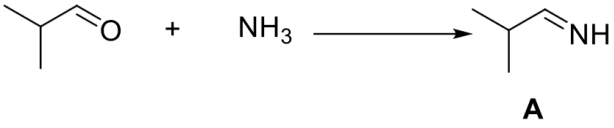
Question 5 (10 marks)

Which structural features does a compound need to have in order to be aromatic? (5 marks) Which of the following compounds are aromatic? Fully justify your answer. (5 compounds, 1 mark each).



Question 6 (20 marks)

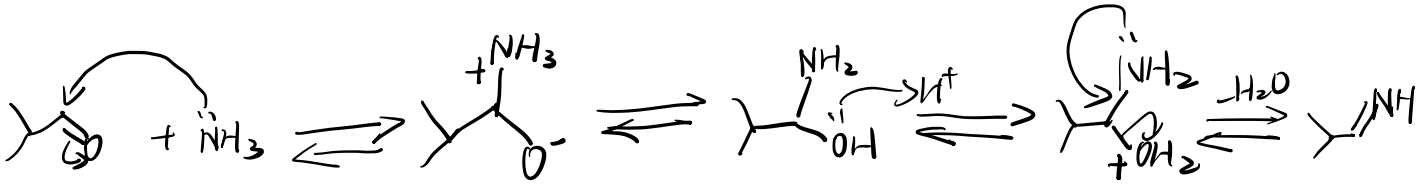
Suggest mechanisms for the reactions that lead to the formation of product **A**. (20 marks)

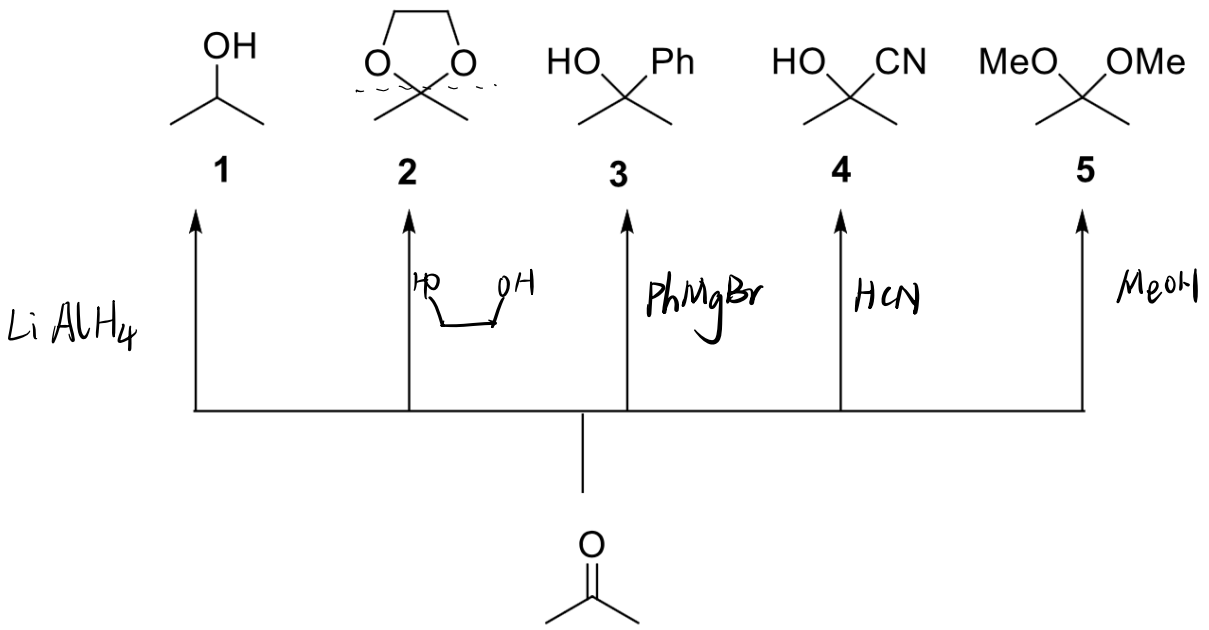


A

Question 7 (10 marks)

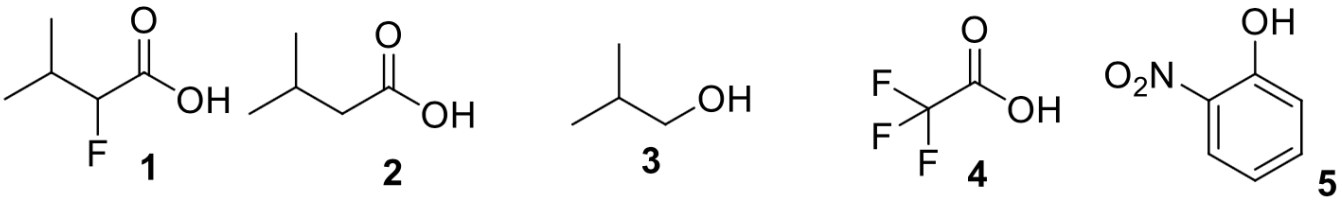
Identify suitable reagents to convert acetone into **1**, **2**, **3, 4** and **5**. No mechanisms required. (2 marks each).





Question 8 (10 marks)

Rank compounds **1** to **5** in order of increasing acidity. Explain your answers.



3 < 5 < 2 < 1 < 4

The End of the Paper

Because acidity : carboxylic acids > phenol > Alcohol , 3 < 5 < 2 .

Inductive effect : fluorine withdraw electron density , stabilizing the conjugate base

Resonance effect : NO_2 can delocalize negative charge through resonance

increasing acidity. 2 < 1 < 4